

SIMATS ENGINEERING
ASSESSMENT TOOL 1: Analytical Problem Solving

COURSE NAME: Cloud Computing and
Big Data Analytics

COURSE CODE: CSA15

COURSE OUTCOME COVERED

CO2: *Analyze virtualization architectures to identify performance and security issues in cloud computing environments. (BL3)*

Assessment Tool Weightage: 40%

Dave's Taxonomy: Manipulation (Level 2)
Question Count: 5

Total Marks: 20

Code of Conduct

I Lochana Sri R.S. /192521348 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Lochana Sri R.S.

Analytical Problem Solving Rubric

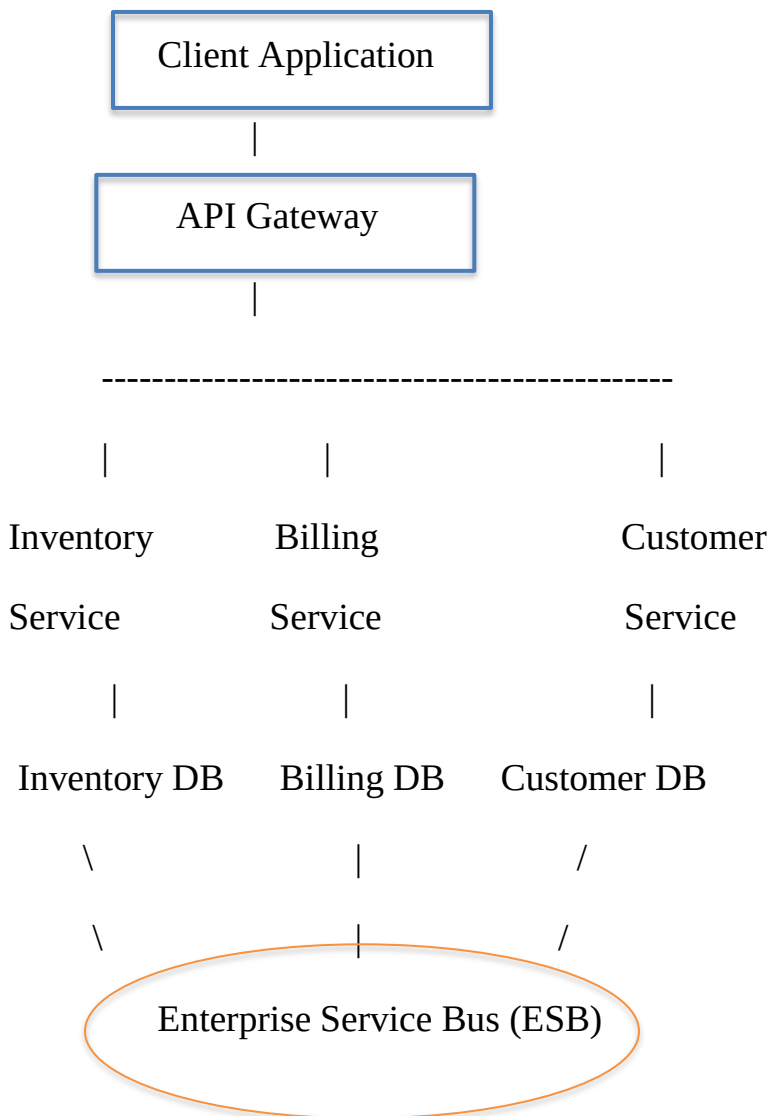
Criteria	Weightage	Level 4 - Exemplary	Level 3 – Proficient	Level 2 - Developing	Level 1 - Beginning
Problem Interpretation	20%	Interprets all technical requirements accurately	Interprets most requirements correctly	Partial interpretation	Misinterprets the problem
Analytical Reasoning	25%	Strong reasoning with clear cause-effect analysis	Good reasoning with minor gaps	Basic analysis only	Weak or no analysis
Method Selection	20%	Chooses highly appropriate virtualization and security concepts	Chooses mostly suitable concepts	Partially suitable concepts	Inappropriate approach
Solution Quality	20%	Provides feasible and technically sound solution	Reasonable solution with minor issues	Basic solution with limitations	Unclear or invalid solution
Presentation of Analysis	15%	Well-structured and technically precise	Generally clear	Somewhat unclear	Difficult to follow

Total Marks Awarded:

Q1. Design an SOA-based solution integrating inventory, billing, and customer management systems.

Introduction

Service-Oriented Architecture (SOA) is an architectural approach where different business functions are developed as independent services that communicate through standard protocols. This enables different systems to work together efficiently while remaining loosely coupled



COMPONENTS:

1. Inventory Service

Functions

- Maintain product details
- Track stock availability
- Update inventory after purchase
- Notify when stock is low

2. Billing Service

Functions

- Generate invoices
- Calculate GST and discounts
- Process payments
- Store billing history

3. Customer Management Service

Functions

- Store customer information
- Manage orders
- Maintain purchase history
- Handle customer support requests

4. Enterprise Service Bus (ESB)

Acts as the communication hub by:

- Routing messages
- Converting data formats
- Coordinating communication among services

WORKING PROCESS:

Step 1

Customer places an order.

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Step 2

Customer Service validates customer information.

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Step 3

Inventory Service checks stock availability.

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Step 4

If stock is available, Inventory Service updates stock quantity.

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Step 5

Billing Service generates invoice and calculates taxes.

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Step 6

Payment is processed.

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Step 7

Order confirmation is sent to the customer.

ADVANTAGES

- Loose coupling between services
- Easy integration of existing applications
- Independent service updates
- Better scalability
- Faster maintenance

- Reusable business services
 - Improved reliability
 - Reduced development cost
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CONCLUSION

An SOA-based solution allows Inventory, Billing, and Customer Management systems to communicate through reusable services. This improves flexibility, scalability, and maintainability while ensuring smooth business operations.

Q2. Evaluate the challenges of implementing SOA in large organizations.

INTRODUCTION

Although Service-Oriented Architecture (SOA) provides flexibility and reusability, implementing it in large organizations is challenging due to the complexity of integrating many systems, departments, and technologies.

CHALLENGES:

1. High Initial Cost

- Requires investment in infrastructure, middleware, ESB, and skilled professionals.
- Existing applications may need modification.

2. Complex Integration

- Organizations often use legacy systems developed with different technologies.
- Integrating all systems into SOA is difficult and time-consuming.

3. Security Issues

- Services exchange sensitive business data over networks.
- Strong authentication, encryption, and authorization are required.

4. Service Governance

- Managing hundreds of services becomes difficult.
- Proper version control, monitoring, and service registry are essential.

5. Performance Overhead

- Communication between multiple services increases network traffic.
- XML/JSON message processing may slow response time.

6. Service Dependency

- Failure of one critical service can affect many dependent applications.
- Backup and fault-tolerance mechanisms are necessary.

7. Skilled Workforce Requirement

- Developers must understand SOA principles, web services, APIs, ESB, and service orchestration.
- Training employees increases implementation cost.

8. Testing Complexity

- Individual services may work correctly, but integration testing across multiple services is complex.
- Continuous testing and monitoring are required.

9. Data Consistency

- Different services maintain separate databases.
- Keeping data synchronized across services is challenging.

10. Organizational Resistance

- Employees may resist changing from traditional monolithic systems.
 - Departments need proper coordination and training.
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POSSIBLE SOLUTIONS

CHALLENGE	SOLUTION
Integration complexity	Use ESB and standardized APIs
Security	Implement encryption, OAuth, and authentication
Performance	Use caching and load balancing
Governance	Maintain service registry and monitoring tools
Legacy systems	Gradually migrate using service wrappers
Skill gap	Provide SOA training programs

ADVANTAGES DESPITE CHALLENGES

- Greater scalability
- Better service reuse
- Easier maintenance
- Faster business process integration
- Improved flexibility
- Lower long-term operational cost

CONCLUSION

Implementing SOA in large organizations can be challenging due to integration, security, governance, and performance issues. However, with proper planning, standardized services, strong governance, and skilled professionals, organizations can successfully implement SOA and gain long-term business benefits